

Simulated free-flight validation — run checklist

Only the identifiable cases fly; S10 and S12 tip under the excitation collective and are excluded by construction. Per trial: set the case offset and mass, configure the feedforward from the *identified* offset (`w_ff` only), arm, take off to the standard hover point, hold ≥ 5 s, stop the log. Same topics as the excitation runs; bag name `<variant>_<trial>` under `S<k>/`.

case	truth [mm]	identified [mm]	m [kg]	<code>wo_ff</code> $\times 3$	<code>w_ff</code> $\times 3$
S1	(−6, +0)	(−6.01, −0.11)	3.066	☐ ☐ ☐	☐ ☐ ☐
S2	(+0, +10)	(−0.01, +10.00)	3.066	☐ ☐ ☐	☐ ☐ ☐
S3	(+10, −5)	(+10.02, −5.05)	3.066	☐ ☐ ☐	☐ ☐ ☐
S4	(+20, +20)	(+20.19, +20.19)	3.066	☐ ☐ ☐	☐ ☐ ☐
S5	(+20, −20)	(+20.22, −20.16)	3.066	☐ ☐ ☐	☐ ☐ ☐
S6	(−20, +20)	(−20.26, +20.09)	3.066	☐ ☐ ☐	☐ ☐ ☐
S7	(−20, −20)	(−20.21, −20.24)	3.066	☐ ☐ ☐	☐ ☐ ☐
S8	(+25, +25)	(+25.22, +25.20)	3.066	☐ ☐ ☐	☐ ☐ ☐
S9	(+32, +32)	(+32.37, +32.09)	3.066	☐ ☐ ☐	☐ ☐ ☐
S11	(+38, +14)	(+38.13, +14.22)	3.066	☐ ☐ ☐	☐ ☐ ☐
S13	(+25, +25)	(+25.35, +25.23)	3.220	☐ ☐ ☐	☐ ☐ ☐

66 flights = 11 cases $\times$ 2 variants $\times$ 3 trials. The identified offsets are the R3 excitation-campaign estimates, frozen in `analysis/freeflight_sim_checklist.py`; the feedforward moments follow as $W\hat{y}$ and $W\hat{x}$.